

MODEL RESEARCH DOCUMENT

Project: AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Introduction:

The AI-Powered Workforce Analytics & Talent Intelligence Dashboard is designed to help organizations convert HR data into actionable business intelligence. The project focuses on workforce performance, employee engagement, attrition analysis, recruitment effectiveness, diversity monitoring, and talent development through an intelligent dashboard interface

The proposed system combines predictive analytics, Large Language Models (LLMs), RetrievalAugmented Generation (RAG), and conversational querying so that HR leaders can monitor trends and make data-driven talent decisions more efficiently [file:1]. The source dataset referenced for this project is the IBM HR Analytics Attrition dataset from Kaggle

Project Objective:

The main objectives of the project are:

To analyze employee and HR data in order to identify patterns related to attrition, engagement, and workforce performance.

To build predictive models that can estimate employee risk, forecast attrition trends, and support workforce planning.

To integrate NLP and LLM capabilities for chatbot-based querying, HR text classification, and employee feedback analysis.

To provide explainable insights so HR managers can understand why a prediction was made and justify interventions with confidence.

To create a scalable dashboard that supports continuous model improvement as workforce conditions evolve.

Key Point:

The most important points in this project are listed below:

The dashboard transforms fragmented HR records into a centralized intelligence platform.

It supports real-time analytics and natural language interaction for faster decision-making.

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Model Research Document Project Title Introduction Project Objectives Key Points The solution balances predictive performance with interpretability, which is important in HR use cases.

Different model families are used for classification, forecasting, segmentation, anomaly detection, NLP, and explainable AI.

Class imbalance handling, feature engineering, and preprocessing significantly affect model quality .

List of model:

The project can use multiple model categories depending on the task.

Model	Main Use in Project	Typical Accuracy Range
Logistic Regression	Baseline attrition prediction with high interpretability	80% - 85%
Random Forest	Robust prediction for non-linear employee data	85% - 89%
XGBoost	High-performance prediction for complex patterns	87% - 92%
Gradient Boosting	Sequential error correction for improved classification	86% - 90%
Support Vector Machine (SVM)	Classification in high-dimensional feature spaces	83% - 87%
Decision Tree	Easy-to-explain attrition decision paths	78% - 83%
Extra Trees	Employee risk prediction with low variance	92% - 96%
TabNet	Interpretable deep learning for tabular HR analytics	93% - 96%
FT-Transformer	High-accuracy structured workforce data modeling	95% - 98%
TabTransformer	Mixed categorical feature learning and interaction modeling	94% - 98%
Isolation Forest	Fraud and outlier detection	High precision
DBSCAN	Employee segmentation and irregular cluster detection	Good

Best NLP/LLM choice for this project :

Among the NLP/LLM options, a combined approach is most suitable for this dashboard. BERT or RoBERTa can be used for employee feedback sentiment analysis and HR text classification, while Sentence-BERT is suitable for semantic search across HR documents and policies

For conversational interaction, Llama 3 or GPT can power a dashboard assistant that answers workforce-related questions in natural language. A practical architecture would use Sentence-BERT for retrieval, an LLM such as Llama 3 or GPT for response generation, and RAG to ground responses in internal HR data.

Model	Use in Dashboard	Typical Accuracy Range
ARIMA	Employee attrition forecasting	82% - 90%
Prophet	Workforce planning and trend forecasting	85% - 92%
LSTM	Long-term workforce trend prediction	90% - 96%

These models help the dashboard move beyond static reporting and support future planning, such as forecasting attrition trends and workforce demand.

Risk and consideration:

Several implementation risks and considerations should be addressed during development:

Class imbalance: Attrition is often the minority class, so methods such as SMOTE or class weighting may be needed to improve prediction quality.

Feature engineering impact: Encoding, scaling, and selection of employee features directly affect model performance.

Interpretability vs performance: Highly accurate models such as XGBoost may be harder to explain than Logistic Regression or Decision Trees, which are easier for HR stakeholders to trust . NLP / LLM Models Best NLP/LLM Choice for This Project Time-Series Forecasting Models Risk and Considerations

Data privacy: HR data contains sensitive employee information, so strict access control and secure data handling are necessary. Bias and fairness: Workforce models may unintentionally reflect bias present in historical data, especially in hiring, performance, or retention decisions. Model drift: Workforce patterns can change over time, so models should be monitored and retrained regularly .

Responsible LLM use: NLP/LLM responses should be grounded in trusted HR data to reduce hallucinations and improve reliability in decision support.

Recommended model strategy:

A strong strategy for would be a layered model design:

Use XGBoost , Random Forest, or FT-Transformer for high-accuracy predictive analytics.

Use Logistic Regression and Decision Tree for interpretable reporting to HR teams.

Use SHAP and LIME to explain risk predictions and important workforce factors.

Use BERT/RoBERTa for text analytics and Sentence-BERT with Llama 3 or GPT for the dashboard assistant.

Use Prophet or LSTM for workforce planning and attrition forecasting.

This combination supports strong prediction, explainability, and interactive intelligence within a single dashboard platform.

Conclusion :

The AI-Powered Workforce Analytics & Talent Intelligence Dashboard is a strategic HR intelligence solution that combines predictive modelling , explainable AI, NLP, LLMs, and timeseries forecasting to improve workforce decision-making .A balanced model selection approach is the most suitable for this project: performance-focused models can drive accurate risk prediction, while interpretable and NLP-enabled models make the system practical, explainable, and useful for HR managers and executives . With proper handling of class imbalance, fairness, privacy, and continuous refinement, the dashboard can become a scalable and effective talent intelligence platform.